

Introduction

Raw RNA-seq read counts depend on library size and library composition; normalisation removes these technical sources of variation so that samples can be compared on equal footing. The standard methods — TMM (edgeR), RLE (DESeq2), upper-quartile, and CPM — differ in how aggressively they handle compositional differences and which assumption they make about the bulk of genes. Choosing the right normalisation is foundational; downstream differential-expression tools incorporate the normalisation factors as offsets in their GLM, so a poor choice undermines every conclusion that follows.

Prerequisites

A working understanding of count data, library size, and the difference between technical (sequencing depth) and biological (true expression) variation between samples.

Theory

The four standard methods:

- **CPM (counts per million)**: simple division of each count by the total library size in millions. Adjusts only for sequencing depth; does not address compositional differences (where one sample has a few highly expressed genes pulling resources from others).
- **TMM (Trimmed Mean of M-values)**: edgeR's default. Computes log-fold-changes between each sample and a reference, trims extreme values, and uses the trimmed mean as the normalisation factor. Robust to highly expressed compositional outliers.
- **RLE (Relative Log Expression)**: DESeq2's `estimateSizeFactors`. Takes the median of the ratio of each gene's count to its geometric mean across samples. Equivalent to TMM in spirit; both adjust for composition.
- **Upper-quartile**: scaling by the 75th-percentile count; less robust than TMM/RLE but useful for very sparse data.

TMM and RLE are the de facto standards; CPM is appropriate only for visualisation, not for statistical comparison.

Assumptions

Most genes are not differentially expressed (the “majority is unchanged” assumption that motivates trimmed-mean and median-based normalisation); samples are technical replicates of comparable biological material.

R Implementation

```
library(edgeR); library(DESeq2)

counts <- matrix(rpois(1000, lambda = 10), 100, 10)
rownames(counts) <- paste0("g", 1:100)

# edgeR TMM
dge <- DGEList(counts = counts)
dge <- calcNormFactors(dge, method = "TMM")
dge$samples$norm.factors

# DESeq2 size factors
dds <- DESeqDataSetFromMatrix(counts, colData = DataFrame(id = 1:10),
                              design = ~ 1)
dds <- estimateSizeFactors(dds)
sizeFactors(dds)
```

Output & Results

`calcNormFactors()` returns per-sample TMM normalisation factors (centred near 1, with deviations indicating compositional differences); `estimateSizeFactors()` returns DESeq2 size factors with similar interpretation. Both are used as offsets in downstream GLMs by edgeR, DESeq2, and limma-voom.

Interpretation

A reporting sentence: “TMM normalisation factors ranged from 0.85 to 1.18 across the 10 samples, with no factor exceeding the conventional $\pm 20\%$ concern threshold; the corresponding DESeq2 RLE size factors agreed within 5 %, confirming the normalisation is robust to method choice.” Quote the range of normalisation factors; extreme values signal compositional anomalies.

Practical Tips

- TMM and RLE adjust for compositional differences; CPM does not. For statistical comparison, always use TMM or RLE; CPM is only appropriate for visualisation.
- Extreme normalisation factors (above 2 or below 0.5) warrant investigation; they often indicate a few very highly expressed genes dominating a sample.
- For single-cell RNA-seq, different normalisation methods apply (scrn’s pooling-based size factors, sctransform’s regularised regression); bulk methods underperform on the sparser, lower-coverage single-cell data.
- Normalisation factors enter the downstream GLM as an offset: $\log(\hat{\mu}_{ij}) = \log(\mathrm{NF}_j \cdot e_{1j}) + X_i^{\text{top}} \beta$. The DE method handles this automatically.
- TMM with `refColumn` set to a specific reference sample is useful when one sample is known to be representative; the default chooses an automatic reference.
- For mixed-tissue or very heterogeneous experiments, the “majority unchanged” assumption can fail; consider quantile normalisation (`limma::normalizeQuantiles`) or condition-aware alternatives.

R Packages Used

`edgeR::calcNormFactors()` for TMM, upper-quartile, and RLE alternatives; `DESeq2::estimateSizeFactors()` for the canonical RLE; `limma::cpm()` and `voom()` for normalisation in the limma framework; `scrn` and `sctransform` for single-cell normalisation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat